

Solved Problem

A silicon PN junction at room temperature and its width are used as an example to solidify understanding of the theoretical derivation.

Problem-1

A silicon PN junction at room temperature (300 K) has a uniform acceptor concentration $N_A = 10^{16} \text{ cm}^{-3}$ in the P-type region and a donor concentration $N_D = 5 \times 10^{16} \text{ cm}^{-3}$ in the N-type region. Assume the built-in potential V_0 is 0.7 V. Calculate the depletion width of this junction.

Constants

- Elementary charge, $q = 1.6 \times 10^{-19} \text{ C}$
- Relative permittivity of silicon, $\epsilon_r = 11.7$
- Permittivity of free space, $\epsilon_0 = 8.854 \times 10^{-14} \text{ F/cm}$
- Permittivity of silicon, $\epsilon = \epsilon_r \epsilon_0$

Solution . Do yourself